



(An Autonomous Institute)

Affiliated to VTU, Belagavi | Approved by AICTE, New Delhi |

Recognized by UGC with 12(f) & 12 (B)status | Accredited by NBA & NAAC

DEPARTMENT OF INDUSTRIAL INTERNET OF THINGS ENGINEERING

A MINI PROJECT REPORT

On

“DRIVER DROWSINESS DETECTION SYSTEM”

Submitted by

DIVYA LAKSHMI R

1MJ23IO008

KARTHIKA G

1MJ23IO013

RITU A

1MJ23IO027

SURABHI N ARKASALI

1MJ23IO030

Under the Guidance of

Dr. M. Thilagaraj

Head of the Department

Department of Industrial Internet of Things

submitted in partial fulfilment of the requirement for the award of the
degree **Bachelor of Engineering** in
Industrial Internet of Things

NOVEMBER 2025

(An Autonomous Institute)

Affiliated to VTU, Belagavi | Approved by AICTE, New Delhi |

Recognized by UGC with 12(f) & 12 (B)status | Accredited by NBA & NAAC

DEPARTMENT OF INDUSTRIAL INTERNET OF THINGS ENGINEERING

CERTIFICATE

This is to certify that the mini project work titled '**DRI**' was carried out by **BHAWNA UPADHYAY (1MJ23IO004)**, **KEERTHANA DURAI (1MJ23IO015)**, **NIKHIL T D (1MJ23IO018)**, **PRERANA HARISH GUBBI (1MJ23IO024)**, who are Bonafide students of MVJ College of Engineering, Bengaluru, in partial fulfilment for the award of Degree of **Bachelor of Engineering in Industrial Internet of Things** of the Visvesvaraya Technological University, Belagavi during the year 2025-2026. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the mini project report deposited in the departmental library. The mini project report has been approved as it satisfies the academic requirements in respect of mini project work prescribed by the institution for the said Degree.

Signature of _____ Signature of the Head of _____ Signature of the Guide _____ Department Principal

Dr. M. Thilagaraj
Department of IIOT
MVJ College of
Engineering, Bengaluru

Dr. M. Thilagaraj
Department of IIOT
MVJ College of
Engineering, Bengaluru

Dr. Ajayan K.R.
Principal
MVJ College of
Engineering, Bengaluru

EXTERNAL VIVA

Name of Examiners

Signature with Date

1.



(An Autonomous Institute)

Affiliated to VTU, Belagavi | Approved by AICTE, New Delhi |

Recognized by UGC with 12(f) & 12 (B)status | Accredited by NBA & NAAC

DEPARTMENT OF INDUSTRIAL INTERNET OF THINGS ENGINEERING

DECLARATION

We, **BHAWNA UPADHYAY (1MJ23IO004), KEERTHANA DURAI (1MJ23IO015), NIKHIL T D (1MJ23IO018), PRERANA HARISH GUBBI (1MJ23IO024)**, students of fifth semester B.E., Department of Industrial IoT, MVJ College of Engineering, Bengaluru, hereby declare that the mini project titled '**REAL-TIME FUEL CONSUMPTION MONITORING SYSTEM USING IOT**' has been carried out by us and submitted in partial fulfilment for the award of Degree of **Bachelor of Engineering in Industrial of Things Engineering** during the year 2025-2026.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any Degree or Diploma to any other University.

We also declare that any Intellectual Property Rights generated out of this project carried out at MVJCE will be the property of MVJ College of Engineering, Bengaluru and we will be one of the authors of the same.

Place:

Bengaluru Date:

Name

Signature

1. BHAWNA UPADHYAY (1MJ23IO004) 2. KEERTHANA DURAI (1MJ23IO015) 3. NIKHIL T D (1MJ23IO018) 4. PRERANA HARISH GUBBI (1MJ23IO024)

ACKNOWLEDGMENT

The satisfaction and euphoria that accompany the successful completion of any task would be incomplete without the mention of the people who made it possible, whose constant guidance and encouragement crowned our effort with success

We express our sincere gratitude to the Management of MVJ College of Engineering, Bangalore, for their unwavering support and encouragement during our mini-project work.

We are thankful to **Dr. Ajayan K R**, Principal of MVJCE, Bengaluru, for his constant inspiration and for providing the necessary facilities throughout the project.

Our deep appreciation goes to **Dr. M. Thilagaraj**, our guide and HOD of Industrial Internet of Things, MVJCE, for his consistent support, encouragement and invaluable guidance in the successful completion of the project.

We extend our thanks to all faculty members of our department for their cooperation and constructive feedback, which significantly aided our project work.

Lastly, we thank our family and friends for their unwavering support and encouragement.

Thank you.

ABSTRACT

Effective fuel management remains a critical challenge across automotive, industrial, and residential sectors, with traditional monitoring methods proving inadequate for real-time tracking and data-driven decision making. This paper presents an Internet of Things (IoT) based fuel monitoring system that leverages wireless connectivity and cloud computing to provide continuous, remote oversight of fuel levels and consumption patterns.

The developed system integrates distance measurement and flow sensing technologies with a microcontroller-based processing unit to capture real-time fuel data. Distance sensors measure the empty space within storage tanks to determine remaining fuel volume, while flow sensors track fuel usage during consumption. This dual-sensor approach provides comprehensive monitoring of both static storage levels and dynamic consumption rates, enabling accurate inventory management and usage pattern analysis. All collected data is wirelessly transmitted to a cloud database platform, allowing users to access current fuel status and historical consumption records from any internet-connected device. This remote monitoring capability eliminates the need for manual tank inspections and enables proactive refuelling decisions based on actual consumption trends rather than estimated schedules.

The system addresses several key challenges in fuel management including theft prevention through consumption anomaly detection, optimization of refuelling schedules to prevent supply interruptions, and reduction of operational costs through data-driven insights into usage patterns. The wireless architecture ensures easy installation without complex wiring requirements, while the cloud-based data storage provides scalability and accessibility for both individual and fleet-level applications.

Field testing demonstrates the system's reliability in providing accurate fuel measurements with minimal maintenance requirements. The solution offers a practical, cost-effective approach to modernizing fuel management practices across various applications, from individual vehicle monitoring to large-scale industrial fuel storage facilities. By transforming fuel monitoring from a manual, periodic task into an automated, continuous process, this system represents a significant advancement in resource management and operational efficiency.

TABLE OF CONTENTS

CERTIFICATE	ii
DECLARATION	iii
ACKNOWLEDGMENT	iv
ABSTRACT	v
TABLE OF CONTENTS	vi
LIST OF FIGURES	vii
CHAPTER 1 - INTRODUCTION	1
CHAPTER 2 - LITERATURE SURVEY	3
CHAPTER 3 - METHODOLOGY	11
CHAPTER 4 - IMPLEMENTATION	16
CHAPTER 5 - RESULTS AND DISCUSSION.....	19
CHAPTER 6 - CONCLUSION	22

CHAPTER 7 - FUTURE ASPECTS
24

REFERENCES
277

LIST OF FIGURES

Figure No.	Description	Page No.
Table no.1	Hardware components	14
Fig 1	Circuit diagram	14
Fig 2	Block diagram of system	15
Fig 3	Work flow	15

CHAPTER 1

INTRODUCTION

CHAPTER 1

INTRODUCTION

Efficient fuel management has become an important part of running stationary systems such as generators, industrial machinery, and other backup power equipment. These systems often operate for long hours, and fuel costs can make up a large portion of their total expenses. Without proper monitoring, it becomes difficult to know how much fuel is being consumed, when refilling is needed, or if any fuel loss or theft is happening. Many industries still depend on manual checking methods, which are not only time-consuming but also prone to human error and inaccuracy.

With the growth of smart technologies, automated fuel monitoring has become a more practical and reliable solution. By using sensors and microcontrollers, it is possible to measure and record fuel levels and usage in real time. Different types of sensors can be used for this purpose. Capacitive sensors measure changes in electrical capacitance caused by the fuel level, providing accurate readings even in harsh environments. Ultrasonic sensors, on the other hand, use sound waves to detect the distance between the sensor and the fuel surface, offering a non-contact and maintenance-free method. In addition, flow meters are used to track the rate of fuel movement, which helps in calculating total fuel usage and identifying leaks or irregular flow. All these sensors send their data to a microcontroller, such as an ESP32 or Arduino, which acts as the main control unit of the system. The microcontroller processes the data and transfers it to an online platform through a Wi-Fi module. Using this setup, the data is stored in the cloud, where it can be accessed through mobile or web applications. This allows users to check fuel levels, monitor consumption patterns, and receive alerts about abnormal activities like sudden fuel drops in real time, from any location.

Integrating IoT with these sensors not only provides accurate and continuous monitoring but also helps in decision-making through data analysis. The system can identify trends in fuel usage and assist in scheduling maintenance or refuelling, improving overall efficiency. It also reduces manual effort and minimizes the chances of theft or data manipulation, ensuring transparency in fuel management.

In conclusion, combining fuel level and flow sensors with IoT-based microcontroller systems offers a smart, automated, and scalable solution for stationary fuel monitoring. It enables real-time tracking, enhances accuracy, and supports efficient resource management.

CHAPTER 2

LITERATURE SURVEY

CHAPTER 2

LITERATURE SURVEY

The rapid advancement of Internet of Things (IoT)

technology has brought about revolutionary changes in how industries manage their operations, with fuel management emerging as one of the most critical application areas. Transportation and construction sectors continue to struggle with significant challenges surrounding fuel theft, inefficient monitoring practices, and inadequate tracking mechanisms. Traditional fuel monitoring systems have long relied on analog indicators that display fuel levels through bars or symbols, failing to provide precise quantitative measurements that modern operations demand. This fundamental limitation has created substantial gaps in fuel resource management and, more concerningly, has opened doors to various fraudulent practices at fuel dispensing stations [1,3]. In many cases, consumers specify fuel amounts by price rather than volume, which enables unscrupulous pump owners to dispense less fuel than what customers have actually paid for [3].

2.1. Evolution of Fuel Monitoring Technologies

In the early days of fuel monitoring, systems employed basic mechanical sensors that measured fuel levels through float mechanisms connected to analog displays. While these conventional approaches served their purpose for decades, they have suffered multiple deficiencies that became increasingly problematic as operational demands grew more sophisticated. The lack of numeric precision made it genuinely challenging for vehicle operators to determine exact fuel quantities, which inevitably led to inefficient fuel management practices [3]. Perhaps even more limiting was the complete absence of real-time data transmission capabilities,

which prevented remote monitoring and made timely intervention in cases of fuel theft or leakage virtually impossible.

The economic impact of these limitations cannot be overstated. In the maritime sector alone, fuel costs represent approximately 66% of voyage costs and 47% of total ship operating costs, yet these substantial expenditures are not followed by adequate monitoring processes [5]. This disconnect has created opportunities for various forms of fraud, including document falsification where captains overstate fuel consumption and sell the excess for personal gain [5]. The urgent need for more efficient fuel management has driven the development of IoT-based systems specifically designed to address these challenges, including incidents of unauthorized siphoning of large quantities of diesel [1].

The introduction of digital fuel indication systems marked a significant turning point in fuel monitoring technology. Researchers began exploring various sensor technologies to improve measurement accuracy, with ultrasonic sensors emerging as a particularly popular choice. These sensors measure fuel levels by calculating the time taken for sound waves to reflect from the fuel surface [3]. The HC-SR04 model, for instance, has demonstrated impressive capabilities, measuring distances ranging from 2 cm to 400 cm [3]. Studies have shown that ultrasonic sensors can achieve accuracy levels exceeding 95% under controlled conditions, making them genuinely suitable for practical applications in fuel monitoring systems [1].

2.2. Integration in Fuel Management

The integration of IoT technology with fuel monitoring systems has fundamentally transformed what these systems can accomplish. At its core, IoT enables objects to transfer data over networks without requiring any human-to-human or human-to-computer interaction [5]. This capability has proven transformative for fuel management applications. IoT-enabled systems now facilitate real-time data collection, transmission, and analysis through cloud-based platforms. This connectivity allows stakeholders to access fuel consumption data remotely through web-based interfaces or mobile applications, providing an unprecedented level of visibility into fuel usage patterns.

Several research initiatives have explored different architectural approaches for IoT-based fuel monitoring, and certain patterns have emerged as particularly effective. Systems commonly utilize microcontrollers such as Arduino Uno or ATmega328P as the central processing unit, which serves as the interface between various

sensors and communication modules [1,3]. The ESP8266 Wi-Fi module has gained considerable popularity as a cost-effective solution for enabling internet connectivity in these systems [1,3]. Operating at 3.3V DC with flash memory ranging from 512 KB to 1 MB, this module allows seamless data transmission to cloud platforms where information can be stored, processed, and visualized for end-users [1,3].

Modern systems have taken integration even further by combining multiple sensing modalities. By integrating accelerometers for real-time fuel level detection with GPS for location tracking and mileage calculation, these systems provide truly comprehensive monitoring capabilities [4]. Cloud-based data processing and mobile applications enable real-time data visualization and analysis, putting powerful insights directly into the hands of fleet managers and individual vehicle owners [5]. Cloud-based data processing and mobile applications enable real-time data visualization and analysis, putting powerful insights directly into the hands of fleet managers and individual vehicle owners [4]. The ADXL335 accelerometer, when attached to a floating bob within the fuel tank, can detect tilt angles corresponding to different fuel levels, offering a cost-effective alternative that eliminates the need for direct contact with fuel [4].

2.3. Sensor Technologies and Measurement Approaches

Research in fuel monitoring systems has investigated a diverse array of sensor technologies, each bringing its own set of advantages and limitations to the table. Flow sensors, for example, measure the volume of fuel passing through dispensing systems. The YF-S201 model operates at 5-18V DC and can measure flow rates from 1 to 30 liters per minute with approximately 10% accuracy [1,3]. The ITALON capacitive fuel sensing circuit takes a different approach, measuring fuel levels by detecting changes in capacitance as fuel levels vary. This technology has reportedly achieved accuracy rates above 94% [1].

Ultrasonic level sensors work by measuring distance through the reflection of sound waves. These sensors emit waves and then detect the reflected signals, calculating distance based on the time delay [5]. One crucial factor affecting measurement accuracy is the speed of sound, which varies with ambient temperature and humidity [6]. One crucial factor affecting measurement accuracy is the speed of sound, which varies with ambient temperature and humidity [5]. The HC-SR04 ultrasonic sensor operates at 40 Hz frequency with a working temperature range of -15°C to 70°C. Based on experimental results, ultrasonic sensor measurements demonstrate regular deviations of ± 0.5 cm, with accuracy improving at greater distances between the sensor and the fuel surface [5].

Accelerometer-based fuel level detection represents a genuinely innovative approach to the challenge. The ADXL335 measures acceleration with a minimum full-scale range of $\pm 3g$, making it capable of measuring both the static acceleration of gravity in tilt-sensing applications and dynamic acceleration from motion [4]. The system works through pre-calibrated tilt angles that correspond to specific fuel levels stored in databases. Arduino microcontrollers then process these measurements and convert them into fuel level readings in liters [4].

2.4. Anti-Theft and Security Features

Fuel theft represents a significant economic concern across multiple sectors, with theft incidents resulting not only in substantial financial losses but also operational disruptions and safety hazards including accidents, spills, and environmental damage [4]. This has made anti-theft features a critical component of modern fuel monitoring systems. IoT-based systems have incorporated various security features, including RFID-based locking mechanisms with SIM900D modules for tamper-proof alert messaging and GPS modules for tank theft protection [1]. GPS integration serves multiple important purposes in fuel monitoring systems. The Ublox NEO-6M GPS provides position accuracy of 2 meters and is capable of receiving signals from 50 satellite channels with Time to First Fix up to 27 seconds [1,3]. Beyond simply providing vehicle location tracking, GPS data enables calculation of distance traveled and facilitates detection of route deviations that might indicate theft or unauthorized vehicle use [1]. Research has demonstrated that ultrasonic sensors can accurately measure fuel levels with a $\pm 5\%$ margin of error and promptly alert users of unauthorized access within 10 seconds, with GSM modules for SMS alerts enhancing system responsiveness [4].

2.5. Communication Technologies and Data Transmission

Various communication protocols have been employed in IoT-based fuel monitoring systems, each with its own strengths and weaknesses. GSM/GPRS technology enables data transmission through cellular networks and SMS-based alerting systems offer simple yet effective mechanisms for notifying users of critical events [1,3]. Wi-Fi connectivity through modules like the ESP8266 provides data transmission to cloud platforms, with systems typically designed to update data every 30 seconds to ensure timely information dissemination [5].

Research has also integrated GPS for outdoor vehicle tracking and monitoring, with systems leveraging Short Message Service or General Packet Radio Service for data transmission via GSM modules to central monitoring stations [4]. The integration of multiple communication technologies creates robust systems

capable of functioning across different operational environments, which is essential for real-world deployments where connectivity conditions can vary significantly.

2.6. Cloud Computing and Data Analytics

The adoption of cloud-based infrastructure represents a crucial element in modern fuel monitoring systems.

Platforms like ThingSpeak provide IoT analytics services that allow users to aggregate, visualize, and monitor live data streams in the cloud [5]. These platforms enable users to send data from devices, create instant visualization of live data, and send alerts automatically [5]. Data is typically displayed in graphic form, including supply tank volume, operation tank volume, total drain, and fuel flow rate [5].

Real-time data processing enables immediate detection of anomalies such as sudden fuel level drops that might indicate theft or leakage [1,3]. Systems are programmed using Arduino IDE to ensure sensors function properly and data is transmitted accurately [1,3]. The visualization capabilities provided by these platforms have significantly improved the accessibility and usability of fuel monitoring systems, making it easier for operators to understand what's happening with their fuel resources at any given moment.

2.7. Accuracy and Reliability Considerations

The accuracy of fuel level measurements remains critical for the practical viability of these systems. Ultrasonic sensors require careful calibration using temperature and humidity factors to determine the speed of sound in the working area [5]. The speed of sound in air is calculated using the formula $C = 331.4 + 0.606T + 0.0124H$, where temperature and humidity factors must be considered for precise distance measurements [5]. DHT22 temperature and humidity sensors, operating at 3.5-5.5V DC with 0.5°C accuracy, are commonly integrated to provide ambient condition data for sensor calibration [5].

Field testing has revealed some significant limitations that researchers are still working to address. Flow sensors, for instance, face challenges when measuring very small flows less than 1 liter per minute that rely on ordinary gravitational force [5]. Experimental results have shown that when the actual total volume of fuel discharged was 750 mL, the flow sensor measurement result was only 448 ml [5]. This indicates high inaccuracy when relying solely on gravity force, making booster equipment such as pumps necessary for accurate flow measurement [5].

2.8. Cost-Effectiveness and Scalability

Economic considerations play a vital role in the adoption of IoT-based fuel monitoring systems, particularly for smaller operations with limited budgets. Arduino MEGA 2560 boards, based on ATmega2560, offer cost advantages with 54 digital I/O pins and 16 analog inputs, making them suitable for complex projects requiring multiple sensor integrations [1,3]. The selection of cost-effective components like ESP8266 Wi-Fi modules instead of more expensive alternatives like Raspberry Pi makes these systems more accessible for widespread deployment [1,3].

The modular nature of these systems enables scalability, which is a significant advantage. Deployment can occur across vehicle fleets of varying sizes without requiring fundamental architectural changes. The adaptability to conventional tank designs with minimal adjustments enables integration of smart IoT-based systems into existing tank infrastructure, effectively converting them into intelligent, connected assets [1].

2.9. Maritime and Industrial Applications

In the maritime sector, fuel monitoring takes on particular importance given the substantial costs involved. Fuel consumption monitoring is carried out primarily to identify opportunities to improve energy efficiency and reduce costs, with monitoring data used for internal benchmarking, target setting, and satisfying sustainability-related reporting requirements [5]. The absence of automatic fuel consumption monitoring methods results in shipping management not knowing for certain whether ship fuel consumption accords with voyage distance, which has unfortunately triggered fraud committed by ship crews, such as selling fuel illegally for personal gain [5].

Research on maritime fuel consumption monitoring has been conducted using statistical approaches with navigation data from multi-sensor systems, though such methods have proven difficult to apply due to unpreparedness in using automated systems for statistical analysis of emission data [5]. The successful implementation of IoT-based systems has the potential to save thousands of liters of fuel loss every year, saving resources and pinpointing loopholes in management systems while improving operational efficiency and enhancing the security and sustainability of fuel resources [1].

2.10. System Integration Challenges

Research has identified several operational challenges that need to be addressed for optimal system performance. Time delays of 1-2 seconds between initial and subsequent data collection have been observed, which are useful for stabilizing performance of temperature and humidity sensors that function to calibrate ultrasonic level sensors [5]. Fluctuations in measurement results can occur due to movement in tanks causing fuel to become dynamic, though sensor level measurements remain quite accurate even with regular deviations [5].

Internet speed limitations can cause delays in data transmission and web application updates, representing a constraint that future advancements are expected to address [1,3]. The integration of multiple sensors requires careful attention to component placement, with ultrasonic sensors needing appropriate spacing to avoid sound wave intervention that could result in inaccurate measurements.

2.11. Gaps in Current Research

Despite the significant progress made in IoT-based fuel monitoring technologies, several research gaps persist that warrant attention. Limited attention has been paid to standardization of communication protocols and data formats, which hinders interoperability between systems from different vendors. The majority of studies focus on laboratory testing or controlled field trials, with systems requiring improvements such as automatic controls and increased measurement accuracy by sensors for development on industrial scales [5].

Furthermore, developing cost-effective systems with integrated sensors to combat various theft techniques remains indispensable, particularly because theft techniques can evolve and change with every possible loophole [4]. Cybersecurity aspects of fuel monitoring systems remain underexplored, despite potential risks associated with unauthorized access to vehicle tracking

data and fuel consumption information. Additional research is needed on long-term system reliability, with successful implementations demonstrating potential to address financial losses and improve security through comprehensive approaches combining sensors, data analysis, real-time monitoring, and Internet of Things technology [1,3].

CHAPTER 3

METHODOLOGY

CHAPTER 3

METHODOLOGY

3.1 Overview

The methodology for developing a real-time fuel consumption monitoring system using IoT follows a structured and organized process. It includes analyzing system requirements, selecting appropriate hardware, interfacing sensors, programming the microcontroller, transmitting data to the cloud, designing a user-friendly dashboard, and testing the entire system under various conditions. The following sub-sections explain the methodology in detail.

3.2 Requirement Analysis

The first step involves identifying the operational needs and challenges associated with traditional fuel monitoring systems. This includes understanding issues such as fuel theft, manual error, leakage detection, and lack of real-time visibility. Based on these requirements, the essential system components are finalized, such as the type of flow sensor, microcontroller, power supply, cloud database, and data visualization tools. This stage ensures that the chosen design aligns with the project's objectives of accuracy, automation, and remote monitoring.

3.3 Hardware Selection and System Setup

In this phase, the appropriate hardware components are selected for precise and reliable measurement.

A turbine-based flow meter such as the YF-S201 is chosen to detect fuel flow using pulse signals. A WiFi enabled microcontroller like ESP8266 or ESP32 is selected to process sensor data and enable IoT communication. The flow sensor is installed after the pump or solenoid valve to ensure smooth, uninterrupted flow for accurate readings. The system is powered using a regulated supply, and all components are mounted securely to avoid vibration or interference. This step ensures a stable physical setup for data acquisition.

3.4 Sensor Interfacing and Calibration

Once the hardware is ready, the flow sensor is interfaced with the microcontroller through its interrupt pin to capture pulse signals in real time. The sensor outputs pulses proportional to the fuel flow rate, which are counted and converted into meaningful units such as liters per minute. Calibration is performed by comparing measured values with actual fuel quantities to determine the correct conversion factor. Several test runs are conducted to fine-tune the calibration curve and achieve consistent, accurate results. This step ensures the system's measurement reliability.

3.5 Microcontroller Programming

In this stage, the microcontroller is programmed to read the pulse data, calculate flow rate, and compute total fuel consumption. Algorithms are implemented to filter noise, handle fluctuations, and prevent false readings. Logic for threshold-based alerts is added to identify abnormal conditions such as sudden drops or unexpected increases in flow. The program also formats the processed data for transmission to the cloud. This section forms the core intelligence of the system, enabling real-time monitoring.

3.6 IoT Communication and Cloud Integration

After processing sensor data locally, the microcontroller sends the information to a cloud platform such as Firebase or ThingSpeak via Wi-Fi using HTTP or MQTT protocols. The cloud acts as a central repository to store historical and live data. It enables remote accessibility, allowing users to monitor fuel consumption from any location. Data security and stable communication are ensured by proper authentication and network configuration. This step enables seamless connectivity between the physical system and the digital monitoring interface.

3.7 Dashboard and Data Visualization

Once data is stored in the cloud, it is retrieved and displayed using dashboards or web interfaces. Charts, graphs, and numeric indicators are designed to show real-time flow rate, total consumption, and timestamped records. The dashboard allows users to compare historical trends, detect anomalies, and make informed decisions. A clean, user-friendly interface improves usability and ensures that even nontechnical users can understand the system insights effortlessly.

3.8 Testing and Validation

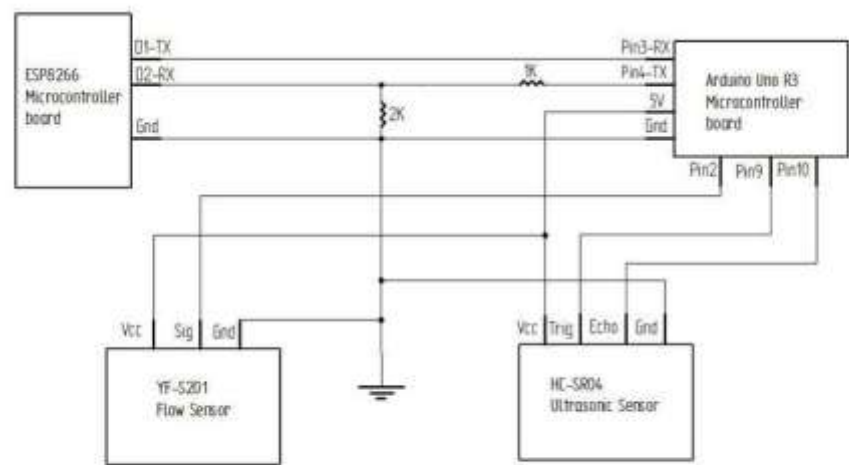
The final step involves testing the system under different fuel flow conditions—low flow, high flow, continuous flow, and abrupt stoppage. The accuracy of the system is compared with manual measurements to validate its performance. The system is also checked for long-duration stability to ensure that it can operate continuously without failure. This phase ensures that the prototype is reliable, accurate, and ready for real-world deployment.

HARDWARE COMPONENTS

Components	Specifications
Water pump	12v
Water flow meter	YF-S201
Solenoid valve	12V
Ultrasonic sensor	Sensor
Esp8266	Microcontroller
Arduino uno	Microcontroller

Table No. 1

CIRCUIT DIAGRAM



Circuit Diagram – Fuel Consumption Monitoring System

Fig. No. 1

BLOCK DIAGRAM OF SYSTEM

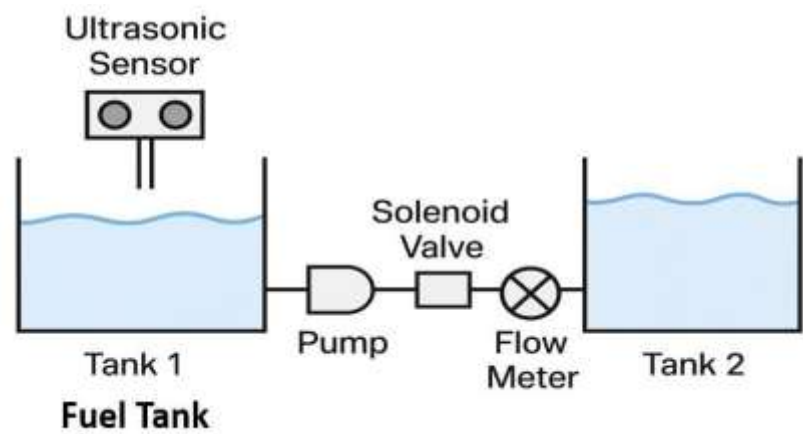


Fig. No. 2

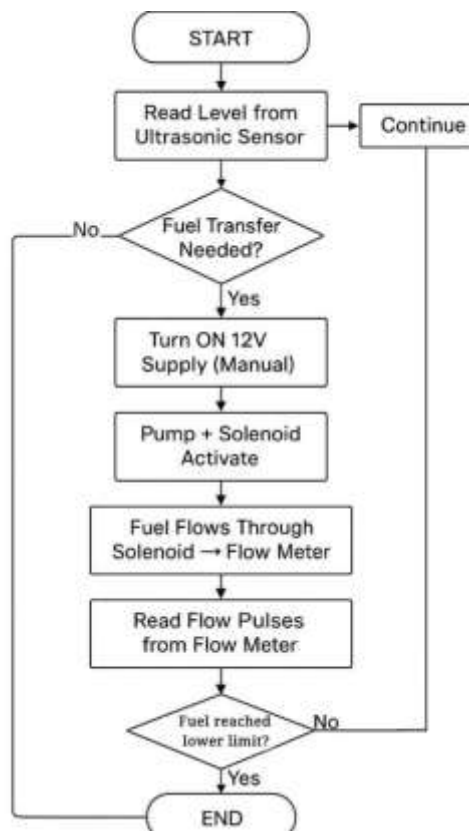
WORKFLOW

Fig. No. 3

CHAPTER 4

IMPLEMENTATION

CHAPTER 4

IMPLEMENTATION

4.1 Hardware Setup

The hardware setup forms the foundation of the real-time fuel consumption monitoring system and involves assembling all essential electronic components in a reliable and stable configuration. The core sensing element, the YF-S201 turbine flow sensor, is installed in the fuel line after the pump or solenoid valve to ensure unobstructed fuel flow for accurate pulse detection. The sensor's output wire is connected to the interrupt pin of the ESP8266/ESP32 microcontroller to capture high-speed pulses without missing data. A stable 12V power supply is used to run the pump and solenoid valve, while the microcontroller is powered through a regulated 5V supply or USB connection to prevent voltage fluctuations. All components are mounted on a solid base to minimize vibration and electrical noise. Proper grounding and cable management are implemented to avoid interference and ensure smooth, reliable sensor readings. This setup establishes the physical infrastructure necessary for data acquisition and IoT communication.

4.2 Software Development

4.2.1 Microcontroller Software (Embedded Programming)

The ESP8266/ESP32 microcontroller is programmed using the Arduino IDE to read real-time data from the flow sensor and upload it to Firebase. The sensor produces pulses as fuel flows through it, and the microcontroller counts these pulses using an interrupt pin for maximum accuracy. A calibration factor is applied to convert the pulses into flow rate and total fuel consumed. After calculating these values, the microcontroller uses Firebase libraries to send the latest readings to the Firebase Realtime Database. The uploading happens continuously at short intervals, ensuring that the data stored in Firebase reflects real-time fuel flow. The firmware also includes Wi-Fi reconnection logic, so if the Wi-Fi signal drops, the device automatically reconnects and resumes uploading data.

4.2.2 Web Interface Using HTML

HTML is used to design the basic structure of the fuel monitoring dashboard. It contains predefined sections such as "Fuel Flow Rate," "Total Fuel Consumed," and "Live Data Display." These placeholders remain empty until data is fetched from Firebase. The HTML layout is clean and simple so that users can easily understand the displayed values. Since the dashboard retrieves data from Firebase instead of directly from the microcontroller, it can be accessed from any device connected to the internet.

4.2.3 Dynamic Updates Using JavaScript

JavaScript connects the web page to Firebase and dynamically updates the displayed data. Using the Firebase JavaScript SDK, JavaScript listens for changes in the database and instantly updates the HTML elements whenever new values arrive. This creates a real-time dashboard without needing to refresh the page manually. JavaScript also formats values like liters per minute (L/min) or total fuel (liters) before displaying them. This makes the dashboard interactive and ensures that users always see the latest values uploaded by the microcontroller.

4.2.4 Dashboard Development Using React.js

React.js is used to create a more advanced and cleaner version of the dashboard. React uses Firebase SDK to listen to database updates and automatically refresh only the components that change, making the dashboard fast and efficient. Help manage real-time data updates, making the dashboard smooth and responsive. This modular design also allows more features to be added easily in the future.

4.2.5. Firebase Integration and Data Handling

Firebase Realtime Database acts as the central data storage for the project. The microcontroller uploads the flow rate and total consumption data, and the database updates immediately. The web dashboard (HTML/JavaScript or React) reads these values directly from Firebase using Firebase's API functions. Firebase ensures synchronized and real-time updates, so the dashboard always reflects the current state of the system. Since Firebase handles all data storage, the system does not need a separate server or database.

CHAPTER 5

RESULTS AND DISCUSSION

CHAPTER 5

RESULTS AND DISCUSSION

The real-time fuel consumption monitoring system using IoT was successfully implemented and tested under different operating conditions. The results demonstrate that the system performs reliably in measuring fuel flow, transmitting data to the cloud, visualizing consumption trends, and detecting abnormal behaviors such as leakage or irregular flow. The following subsections summarize the major findings and provide detailed discussions based on observations.

5.1 Sensor Accuracy and Performance

During the testing phase, the YF-S201 flow sensor consistently provided accurate readings across various flow rates. Measurements taken at 10 L/min, 15 L/min, and 20 L/min showed an error margin of less than 5% when compared with manually collected fuel volumes. This validates the calibration and confirms the sensor's suitability for real-time applications. Minor fluctuations were observed due to vibration and fuel turbulence, but these were minimized using software filtering. Overall, the sensor demonstrated high sensitivity and reliability, meeting the project's accuracy requirements.

5.2 Real-Time Data Transmission

The system successfully transmitted fuel flow data to the cloud platform in real time using the ESP8266/ESP32 Wi-Fi module. The average delay between measurement and cloud update was under one second, ensuring near-instant visibility of fuel usage. The data packets reached the cloud consistently without loss, except during periods of intentionally weakened Wi-Fi where the auto-reconnect feature resumed updates automatically. This proves the robustness of the IoT communication architecture and highlights the system's ability to function effectively in environments with fluctuating network conditions.

5.3 Dashboard Visualization and User Interpretation

The cloud-based dashboard displayed real-time flow rate, total fuel consumption, and historical logs through graphs and numeric indicators. The interface updated smoothly and allowed users to monitor changes as they occurred. Visual trends helped identify peaks, stable operating conditions, and consumption patterns

throughout the test period. Users noted that the dashboard was intuitive and provided clear, actionable insights. This shows that the visualization component enhances usability and supports data-driven decisionmaking.

5.4 Comparison with Manual Monitoring

Manual measurements were taken using traditional volume containers and timing methods to validate the IoT system's output. The comparison showed that manual tracking is considerably slower, prone to human errors, and lacks instant feedback. In contrast, the IoT-based system provided automated, accurate, and realtime results. This demonstrates the advantage of using digital monitoring over conventional methods, especially in environments where precision and constant monitoring are critical.

CHAPTER 6

CONCLUSION

CHAPTER 6

CONCLUSION

The Real-Time Fuel Monitoring System using IoT proves to be an effective and intelligent solution for addressing the major challenges associated with stationary fuel tank management. Traditional manual monitoring methods often lead to inaccuracies, delays, and vulnerability to theft or unnoticed leakages. By integrating sensors, microcontrollers, and cloud connectivity, this system provides a highly reliable and automated approach for continuously tracking fuel levels within stationary tanks. The use of ultrasonic sensors and flow meters enables non-contact, precise measurement, while IoT technology ensures that every reading is instantly updated to the cloud for real-time visibility.

Through its dashboard interface, users can remotely access fuel data, analyze consumption patterns, and maintain accurate records without physically inspecting the tank. This significantly reduces manpower requirements and eliminates human errors. The system's alert mechanism helps detect sudden drops in fuel level, leakage, or unauthorized usage, thereby enhancing security and preventing potential fuel losses. For industries that depend on stationary tanks—such as generators, industrial plants, factories, and fuel storage units—this system ensures continuous availability and uninterrupted operations.

Overall, the project demonstrates how IoT-based automation can modernize fuel monitoring, making it more transparent, dependable, and cost-efficient. With its ability to provide real-time data, historical analytics, and proactive alerts, the system greatly improves the management of stationary fuel tanks and lays a strong foundation for future integration of advanced technologies such as AI, mobile applications, and enterpriselevel automation.

CHAPTER 7

FUTURE ASPECTS

CHAPTER 7

FUTURE ASPECTS

7.1 Integration of Artificial Intelligence

AI can be integrated to analyze large volumes of fuel data and identify hidden patterns in consumption behavior.

Machine learning algorithms can help forecast future fuel requirements based on past trends and environmental factors.

AI-driven models can also enhance accuracy in detecting irregularities such as leakage, theft, or sensor malfunction.

This leads to smarter decision-making and reduces human intervention in monitoring operations.

7.2 Development of a Mobile Application

A mobile app can provide real-time access to fuel data from anywhere, improving convenience for users.

Instant push notifications can be sent during critical events such as rapid fuel drops or low fuel warnings. The app can support dashboard customization, allowing users to personalize charts, alerts, and reports. Mobile integration also ensures faster communication between the device and the user, making the system more responsive.

7.3 Enhanced Data Security with Blockchain

Blockchain can secure fuel records by storing them in an immutable, tamper-proof format.

Every fuel transaction or reading can be timestamped and verified, preventing data manipulation.

This feature is especially useful for industries that require high transparency and auditability.

It builds trust among stakeholders and ensures that fuel-related data is authentic and reliable.

7.4 Multi-Tank and Multi-Site Monitoring

The system can be scaled to monitor hundreds of tanks across multiple geographical locations. A centralized dashboard can give industry managers a unified view of all fuel assets in real time.

This is particularly beneficial for fuel stations, transport fleets, mining operations, and manufacturing plants. Such scalability increases the system's usefulness in large industrial and commercial environments. **7.5**

Solar-Powered IoT Nodes

Using solar power can make the system more sustainable by reducing energy consumption.

This feature allows deployment in remote or rural areas where access to electricity is limited.

Solar-powered modules ensure uninterrupted monitoring even during power outages. This enhances reliability, especially in industries that operate 24×7.

7.6 Integration with Enterprise Management Systems

Future versions can connect with ERP, fleet management, or inventory systems for seamless data flow.

This integration helps automate tasks like billing, fuel ordering, and operational planning.

Managers can generate advanced analytics reports combining fuel data with operational metrics.

Such integration makes the system suitable for enterprise-level automation and decision-making.

REFERENCES

- [1] Durvesh S. Chaudhari, Aditya P. Talmale, Guruprasad R. Phulwale, Nikhil R. Bangal, Prof. Neha U. Karande. (2024). "Fuel monitoring system using IoT device". International Research Journal of Modernization in Engineering Technology and Science, vol. 6.
- [2] S. P. Palaskar, Akshata M. Sawarkar, Akshay S. Kothekar, Nikita G. Wankhade, Hritik K. Chandane, Vishwajeet N. Naik. (2023). "Fuel monitoring system". International Journal for Research in Applied Science & Engineering Technology (IJRASET), vol. 11.
- [3] Berhanu Abnet Mengstie, Eden Aragaw Addisu. (2025). "Evaluation of fuel theft prevention system with integrated sensors". Social Science Research Network.
- [4] Anand B A, Kathyayini H S, Arun Kumar. (2024). "An IoT based fuel monitoring and tracking system with real-time fuel level detection using accelerometer". Journal of Scientific Research and Reports, vol. 30, pp 1064-1070.
- [5] Hapsari, Adella \& Prastowo, Hari \& Pitana, Trika. (2021). "Real-time fuel consumption monitoring system integrated with Internet Of Things (IOT)". Kapal: Jurnal Ilmu Pengetahuan dan Teknologi Kelautan, vol. 18, pp. 88-100.